

Introduction to Geogebra

1. Download Geogebra (Google it, you'll find it). It's free.
2. Open Geogebra. It will open with coordinate axes in the graphics window, which we don't want. Right click anywhere on the Graphics window and click on Axes. They will disappear.
3. At the top of the screen are several buttons that activate different tools. The lower right hand corner of each button will give you a drop down that can change what the button does.
 - a. Notice that if you hover over a button, you get a little summary of how to use the button. For example, if you hover over the Perpendicular Line button, it says "Select point and perpendicular line." This tells you that to use this button you need to first select a point, then select a line. You will get a line through the selected point that is perpendicular to the selected line. These little summaries are very handy when you forget the syntax of how a particular button works.
4. Your first task is to play around with the tools and figure out how they work. You should be able to:
 - a. Put a point on the screen (button 2) and drag it around (button 1).
 - b. Create a line segment, line, and ray (button 3). You should be able to drag around the endpoints.
 - c. Create a midpoint, intersection point, and a point on an object. (button 2)
 - d. Create a perpendicular line, parallel line, perpendicular bisector, and angle bisector (button 4)
 - e. Create a polygon (button 5)
 - f. Create a circle with center through point (button 6)
 - g. Button 7 is fairly worthless for this class
 - h. Measure an angle and measure a distance (button 8). Notice that when you measure an angle, the order in which you select the points matters. If you do a few, you'll figure it out.
 - i. The transformations in button 9 are cool and you can play around with them if you want to. We'll do a lot with them but that is another lesson. For now, you can just focus on the basics
 - j. Be able to put text on the screen (button 10). If you right click the text box you can change the font size.
 - k. The most useful thing about button 11 is the ability to drag the whole screen around. Try it.
5. As you were doing the above exercise, you probably noticed that a list of the objects you created appeared to the left in the "algebra" window.
 - a. Right click on any object, either in the graphics window or the algebra window. One of the choices will be "Show Object". If you click on that, the object will disappear from the screen, but it will still exist. (You'll figure out what that means after you play with this a while). You can reverse this process by right clicking on the object in the Algebra window.

Assignment to turn in is on the back.

Turn this in.

6. This is what you need to turn in.
 - a. Create a random triangle on the screen. Measure all of the angles.
 - i. The angle measuring tool takes some getting used to. The order in which you select the points matters. It's point-vertex-point, but it's hard to remember whether you need to go clockwise or counter-clockwise. If you select the points in the wrong order, it measures a different angle from what you thought it would. Play with this and you'll see what I mean. I can never remember if it's clockwise or counter-clockwise so I just measure it one way, and then if that doesn't give me what I wanted, I sigh deeply, hit the "undo" button, and do it the other way.
 - b. Notice that Geogebra will name the angle measurement with a Greek letter. Now go down to the "Input" field at the bottom of the screen. If you click in the field (like you're going to type something), an icon will appear on the right that will let you enter Greek characters.
 - c. Use this to calculate the sum of any two of the angles in your triangle. Note that this sum appears in the "algebra" window, but you can click and drag it onto the "Graphic" window. Drag this sum underneath your triangle.
 - d. Now create the sum of all three angles. Drag this under your triangle.
 - e. Drag around the points of the triangle and notice what happens to the two sums. Create a text box (Insert Text) on the screen and write a brief summary of what you noticed about the two sums (one two-angle sum, and one three-angle sum). These don't need to be very profound, just simple things like "it changed" or "it didn't change" will do.
 - f. Create a text box somewhere and type in your name.
 - g. Print this and turn it in.